ECO-LOOP BUILDING AGENTS

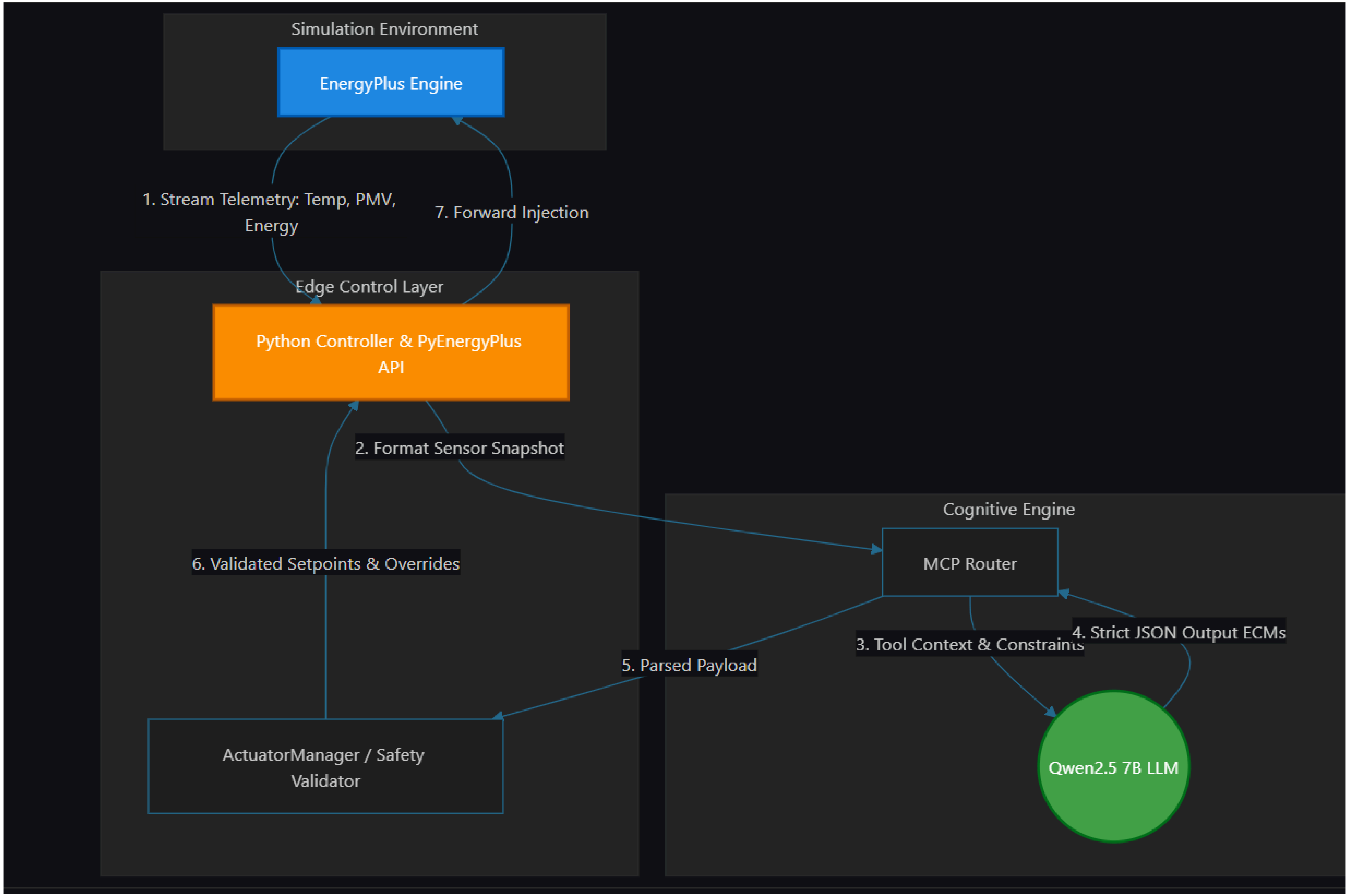
Problem Statement ID – **1**
Problem Statement Title – **Autonomous Smart Building Optimization**
Theme – **Energy & Sustainability / Smart Automation**
PS Category – **Software**
Student Name – **Hari Om**
Student ID – **20472445**

ECO-LOOP BUILDING AGENTS

Proposed Solution:

- A fully autonomous, closed-loop Building Management System (BMS) proof-of-concept pairing a physics-based simulation engine (EnergyPlus) with a locally-run open-source LLM (Qwen2.5 7B) to dynamically adjust HVAC and lighting setpoints in real time
- Replaces static, non-adaptive BMS schedules with a self-correcting agent that continuously balances human comfort (PMV) against aggressive energy reduction
- 100% edge-deployable: runs fully offline with zero cloud dependency, for maximum data privacy and zero API latency
- Features a "Zero-Occupancy Eco Mode" that aggressively cuts energy use when the building is empty, and a "Free Cooling" logic that maximizes outside air when conditions allow

FLOW OF THE SYSTEM



TECHNICAL APPROACH

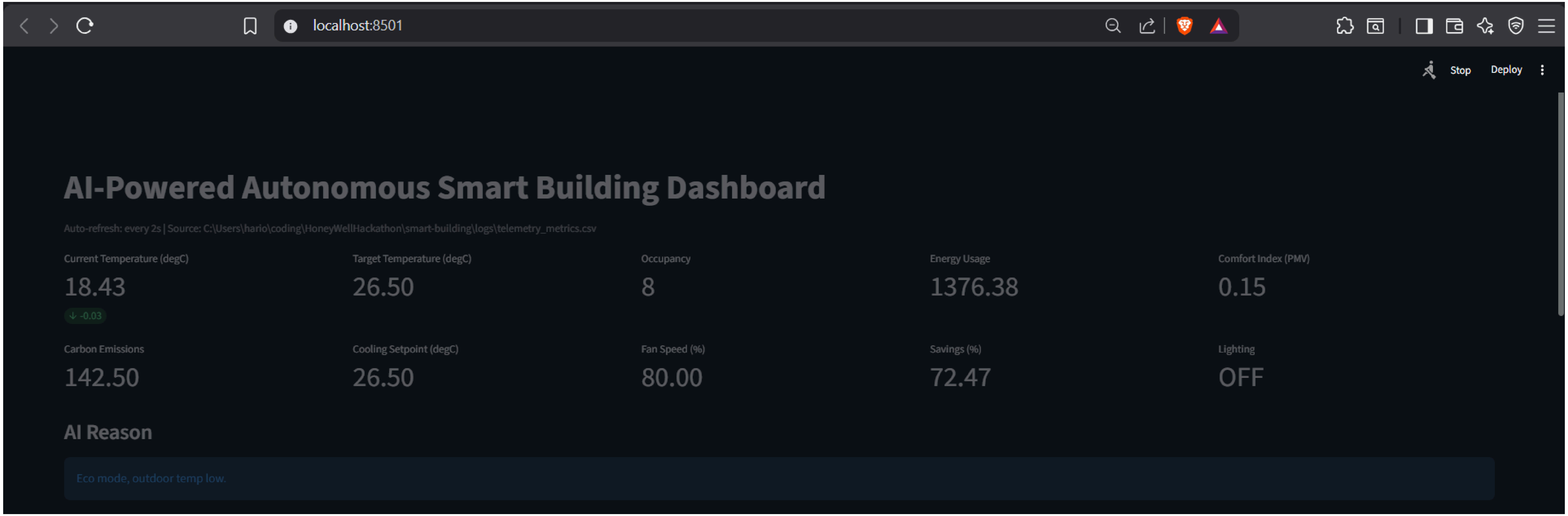
- Simulation Engine: EnergyPlus via the PyEnergyPlus API
- Cognitive Engine: Qwen2.5 (7B parameters), running locally via Ollama
- Architecture: Model Context Protocol (MCP) decouples the LLM from Python control execution
- Dashboarding: Streamlit, Pandas and Plotly for real-time quantitative savings visualization
- Methodology: (1) EnergyPlus streams live telemetry (Temp, PMV, Energy) → (2) Python Controller builds a Sensor Snapshot → (3) MCP Router passes context and safety constraints to the LLM → (4) LLM returns strict JSON Energy Conservation Measures (ECMs) → (5) ActuatorManager validates setpoints against safety limits → (6) Python Controller injects approved overrides back into live EnergyPlus memory

FEASIBILITY AND VIABILITY

- Challenge: LLM hallucinations (e.g. invalid data types) risk crashing the simulation → Mitigation: Pydantic-based fallback loops via a custom LLMOutputParser that intercepts and coerces malformed output before it reaches the simulator
- Challenge: Missing actuator handles in standard baseline .idf files → Mitigation: A resilient ActuatorManager that gracefully skips unavailable variables (e.g. fan speed) and falls back to alternative measures (e.g. cooling setpoints) without breaking the control loop
- Challenge: Prompt latency in a real-time BMS → Mitigation: A quantized 7B model hosted locally to eliminate network round-trips, with generation bounded (max_tokens: 150) for synchronous forward injection

ARTIFACTS

Real-time LLM reasoning trace mapped against live sensor telemetry — showing how each Energy Conservation Measure is derived



ARTIFACTS

Comparing static baseline schedules against the AI-driven controller quantifying the energy savings and comfort trade-offs achieved

ARTIFACTS

Time-series trends for temperature, PMV comfort index, and cumulative energy consumption across the simulation run

RESEARCH AND REFERENCES

- GitHubRepository: https://github.com/hariomkhonde108/HoneyWellHackathon